

Kapture Finance Collections Voicebot

High-Level Design (HLD) — AI Delivery Intern Take-Home Assignment

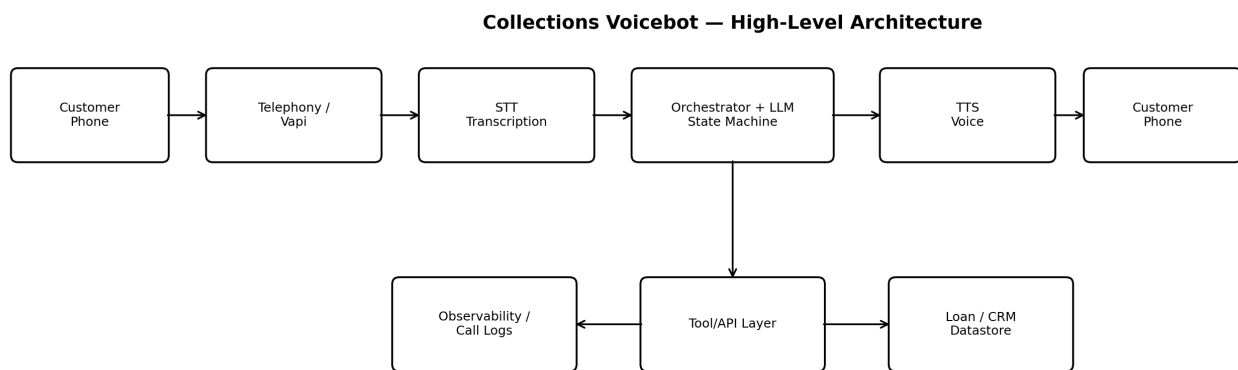
Scenario: Outbound collections agent “Maya” calls a customer with an overdue personal-loan EMI. Example context: Rahul Sharma, overdue EMI ₹8,499, 12 days past due.

Design objective: Authenticate the customer, disclose the purpose and overdue amount only after verification, understand intent, obtain a payment commitment where appropriate, or route the call safely.

1. Architecture & Pipeline

The proposed pipeline is telephony → speech-to-text → stateful orchestrator/LLM → tools/APIs and datastore → text-to-speech → telephony. The state machine is the control layer around the LLM; the LLM cannot independently bypass authentication.

Architecture diagram:



Critical control: debt-related disclosure and collection tools are unavailable until auth_state = VERIFIED

Target conversational latency budget: ~1.5–2.5s for a normal turn; tool/API calls are bounded and logged

Target latency budgets (engineering targets, to be measured during testing):

Hop	Target budget	Purpose
Telephony / Vapi	≤ 250 ms	Audio transport and session handling
STT	≤ 500 ms	Partial/final transcription
Orchestrator + LLM	≤ 800 ms	Intent/state decision and response generation
Tool/API	≤ 400 ms	Verification, account lookup, disposition, etc.
TTS	≤ 500 ms	First audio generation
Target conversational turn	~1.5–2.5 s	Overall perceived response time

2. Conversation Flow / State Machine

CALL_INIT: confirm the outbound session is active and apply calling-policy checks.

DISCLOSURE: Maya identifies herself/company and gives a neutral purpose statement without revealing debt details.

AUTH_PENDING: request the minimum approved verification information. No debt amount, overdue status, loan type, or collection details are disclosed.

AUTHENTICATED: only this state unlocks debt-related account tools and disclosure.

INTENT: classify the customer's response as will-pay, cannot-pay/hardship, dispute, already-paid, callback, wrong-person, hostile, or do-not-call.

RESOLUTION: collect a payment commitment where appropriate; otherwise provide the approved next step or escalate.

DISPOSITION: every completed call writes a final outcome such as PTP, hardship, dispute, already-paid, wrong-person, callback, DNC, escalated, or failed contact.

State lock: if auth_state != VERIFIED, the system must reject debt-related disclosures and payment-collection tools even if the model attempts to invoke them.

3. Intents & Entities

Intent	Key entities / extraction	Action
WILL_PAY	PTP date, amount, payment method if offered	Confirm commitment; log PTP
CANNOT_PAY / HARDSHIP	Reason, requested callback/date	Follow approved hardship flow or escalate
DISPUTE	Dispute reason, disputed amount	Do not argue; escalate / create case
ALREADY_PAID	Payment date, reference if provided	Acknowledge; verify via tool; disposition
WRONG_PERSON	Relationship / callback details if volunteered	Do not disclose debt; end safely
CALLBACK_REQUEST	Preferred date/time	Log callback
HOSTILE	None required	De-escalate; offer human handoff / end
DO_NOT_CALL	None required	Record opt-out and stop collection conversation

4. Tools / API Contracts

Tool	Inputs	Outputs / effect
get_account_details	customer_id	Account status, loan type, overdue amount; callable only after verification
verify_customer	customer_id, verification fields	verified=true/false; reason code
log_promise_to_pay	customer_id, amount, PTP date	PTP ID, status
send_payment_link	customer_id, amount	Link ID/status; trigger approved SMS/WhatsApp mock
escalate_to_agent	customer_id, reason, summary	Escalation ID/status
mark_disposition	call_id, disposition, summary	Logged outcome/status

5. Authentication & Data Safety

The bot follows a verification-before-disclosure policy. A person answering the phone is not assumed to be the customer. Until verification succeeds, Maya uses neutral language and does not reveal the customer's overdue EMI, amount, loan type, or other debt details.

The backend should enforce the lock independently of the prompt: debt-related tools require a verified session token/state. Tool responses should return only the minimum data required for the current state. Call logs should avoid unnecessary sensitive data and should store structured outcomes rather than raw secrets.

6. Guardrails & Compliance

Mandatory controls include self/company/purpose disclosure; calling-policy checks before initiating the call; no threats, harassment, intimidation, or fabricated consequences; immediate handling of do-not-call requests; no disclosure to third parties; no invented account facts; and safe handling of off-topic or adversarial requests.

7. Edge Cases

Already paid → acknowledge and verify using the account tool; do not pressure the customer while the status is unresolved.

Dispute → capture the reason, avoid arguing, and escalate/create the approved case path.

Wrong person → do not reveal the debt; end without confirming why the call was placed.

Do-not-call → acknowledge, record the opt-out/disposition, and stop the collection flow.

Voicemail/no input → use bounded retries/timeouts and end with an appropriate disposition.

Hostile caller → remain calm, do not retaliate, offer the approved escalation/end path.

Language switch EN/Hi → detect the requested language and continue the same state machine and guardrails in the selected language.

8. Escalation & Disposition

Escalate when the customer requests a human, presents a dispute or hardship case outside the automated flow, or when a tool/system failure prevents safe resolution. The call must finish with `mark_disposition` so every attempt has a structured outcome.

9. Observability

Log call/session ID, state transitions, authentication result, intent, tool name/status/latency, disposition, transfer reason, and error category. Track containment rate, PTP rate, average latency, call drop rate, tool error rate, authentication failure rate, and edge-case handling success.

10. Assumptions

Calling hours, verification fields, payment-link channel, escalation queue, and production compliance rules are configurable policy inputs. The take-home uses mocked APIs where needed. The architecture is designed so these policy values can be changed without rewriting the core conversation logic.